

TIME SERIES ANALYSIS & CLASSIFICATION

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Chapter VI: Model Estimation

1. Method of Moments
2. Least Squares Estimation
3. Maximum Likelihood Estimation (MLE)

Time Series Modeling

- ▶ We learned about how to specify our time series model (decide which specific model to use).
- ▶ The general model we have considered is the $ARIMA(p, d, q)$ model.
- ▶ The simpler models like AR, MA, and $ARMA$ are special cases of this general $ARIMA(p, d, q)$ model.

Time Series Modeling

- ▶ Now assume we have chosen appropriate values of p , d , and q (possibly based on evidence from the ACF, PACF, ... etc).
- ▶ We now must estimate the unknown parameters in that stationary $ARMA(p, q)$ model.

We will discuss three estimation techniques: method of moments, least squares, and maximum likelihood.

METHOD OF MOMENTS

Method of Moments

Terminology: The method of moments (MOM) approach to estimation consists of equating sample moments to the corresponding population (theoretical) moments and solving the resulting system of equations for the model parameters.

MOM: Autoregressive Models ($AR(1)$)

- Consider the stationary $AR(1)$ model

$$Y_t = \phi Y_{t-1} + e_t,$$

where $\{e_t\}$ is zero mean white noise with $\text{var}(e_t) = \sigma_e^2$.

- In this model, there are two parameters: ϕ and σ_e^2 . The MOM estimator of ϕ is obtained by setting the population lag one autocorrelation ρ_1 equal to the sample lag one autocorrelation r_1 and solving for ϕ , that is,

$$\rho_1 = r_1.$$

- For this model, we know $\rho_1 = \phi$. Therefore, the MOM estimator of ϕ is

$$\hat{\phi} = r_1.$$

MOM: Autoregressive Models ($AR(2)$)

- For the $AR(2)$ model,

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + e_t,$$

there are three parameters: ϕ_1 , ϕ_2 , and σ_e^2 .

- To find the MOM estimators of ϕ_1 and ϕ_2 , recall the Yule-Walker equations for the $AR(2)$:

$$\begin{cases} \rho_1 = \phi_1 + \rho_1 \phi_2 \\ \rho_2 = \rho_1 \phi_1 + \phi_2. \end{cases}$$

MOM: Autoregressive Models ($AR(2)$)

- ▶ Setting $\rho_1 = r_1$ and $\rho_2 = r_2$, we have

$$\begin{cases} r_1 &= \phi_1 + r_1\phi_2 \\ r_2 &= r_1\phi_1 + \phi_2. \end{cases}$$

- ▶ Solving this system for ϕ_1 and ϕ_2 produces the MOM estimators

$$\hat{\phi}_1 = \frac{r_1(1 - r_2)}{1 - r_2}, \quad \hat{\phi}_2 = \frac{r_2 - r_2^2}{1 - r_2}.$$

MOM: Autoregressive Models

- For the general $AR(p)$ process,

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots + \phi_p Y_{t-p} + e_t,$$

there are $p + 1$ parameters: $\phi_1, \phi_2, \dots, \phi_p$ and σ_e^2 .

- We recall the Yule-Walker equations:

$$\left\{ \begin{array}{l} \rho_1 = \phi_1 + \phi_2 \rho_1 + \phi_3 \rho_2 + \cdots + \phi_p \rho_{p-1} \\ \rho_2 = \phi_1 \rho_1 + \phi_2 + \phi_3 \rho_1 + \cdots + \phi_p \rho_{p-2} \\ \vdots \\ \rho_p = \phi_1 \rho_{p-1} + \phi_2 \rho_{p-2} + \phi_3 \rho_{p-3} + \cdots + \phi_p \end{array} \right.$$

MOM: Autoregressive Models ($AR(p)$)

- ▶ We set $\rho_1 = r_1, \rho_2 = r_2, \dots, \rho_p = r_p$ to obtain

$$\begin{cases} r_1 &= \phi_1 + \phi_2 r_1 + \phi_3 r_2 + \dots + \phi_p r_{p-1} \\ r_2 &= \phi_1 r_1 + \phi_2 + \phi_3 r_1 + \dots + \phi_p r_{p-2} \\ &\vdots \\ r_p &= \phi_1 r_{p-1} + \phi_2 r_{p-2} + \phi_3 r_{p-3} + \dots + \phi_p \end{cases}$$

- ▶ The MOM estimators $\hat{\phi}_1, \hat{\phi}_2, \dots, \hat{\phi}_p$ solve this system of equations.

MOM: Autoregressive Models ($AR(p)$)

- Note that this is a well-posed system (with a square coefficients matrix $\mathbf{R}$), i.e., with the same number of constraints (equations, $\mathbf{R}$'s rows) as unknowns (the elements ϕ_j of the unknown vector Φ). Further, $\mathbf{R}$ is full-rank and symmetric, so that invertibility is guaranteed,

$$\Phi = \mathbf{R}^{-1}\rho$$

MOM: Moving Average Models ($MA(1)$)

- ▶ Consider the invertible $MA(1)$ process $Y_t = e_t - \theta e_{t-1}$, where $\{e_t\}$ is zero mean white noise with $\text{var}(e_t) = \sigma_e^2$.
- ▶ In this model, there are two parameters: θ and σ_e^2 .
- ▶ To find the MOM estimator of θ , we solve

$$\rho_1 = \frac{-\theta}{1 + \theta^2} = r_1,$$

which gives

$$r_1\theta^2 + \theta + r_1 = 0.$$

MOM: Moving Average Models

- ▶ Using the quadratic formula, we find that the solutions to this equation are

$$\theta = (-1 \pm \sqrt{1 - 4r_1^2})/2r_1.$$

- ▶ If $|r_1| > 0.5$, then no real solutions for θ exist.
- ▶ If $|r_1| = 0.5$, then the solutions for θ are ± 1 , which corresponds to an $MA(1)$ model that is not invertible.
- ▶ If $|r_1| < 0.5$, the invertible solution for θ is the MOM estimator

$$\hat{\theta} = (-1 + \sqrt{1 - 4r_1^2})/2r_1.$$

MOM: Moving Average Models

- For higher order *MA* models, the difficulties become more pronounced. For the general $MA(q)$ case, we are left to solve the highly nonlinear system

$$r_k = \begin{cases} \frac{-\theta_k + \theta_1\theta_{k+1} + \dots + \theta_{q-k}\theta_q}{1 + \theta_1^2 + \theta_2^2 + \dots + \theta_q^2}, & k = 1, 2, \dots, q-1 \\ \frac{-\theta_q}{1 + \theta_1^2 + \theta_2^2 + \dots + \theta_q^2}, & k = q \end{cases}$$

for $\theta_1, \theta_2, \dots, \theta_q$. Just as in the $MA(1)$ case, there will likely be multiple solutions, only of which at most one will correspond to a fitted invertible model.

- **IMPORTANT:** MOM estimates are not recommended for use with MA models. They are hard to obtain and they are not necessarily good estimates.

MOM: White Noise Variance

- We now wish to estimate the white noise variance σ_e^2 . To do this, we first note that for any stationary ARMA model, the process variance $\gamma_0 = \text{var}(Y_t)$ can be estimated by the sample variance

$$S^2 = \frac{1}{n-1} \sum_{t=1}^n (Y_t - \bar{Y})^2.$$

MOM: White Noise Variance

- For a general $AR(p)$ process, we recall that

$$\gamma_0 = \frac{\sigma_e^2}{1 - \phi_1\rho_1 - \phi_2\rho_2 - \cdots - \phi_p\rho_p}$$
$$\Rightarrow \sigma_e^2 = (1 - \phi_1\rho_1 - \phi_2\rho_2 - \cdots - \phi_p\rho_p)\gamma_0.$$

- Therefore, the MOM estimator of σ_e^2 is obtained by substituting in $\hat{\phi}_k$ for ϕ_k , r_k for ρ_k , and S^2 for γ_0 . We obtain

$$\hat{\sigma}_e^2 = (1 - \hat{\phi}_1 r_1 - \hat{\phi}_2 r_2 - \cdots - \hat{\phi}_p r_p) S^2.$$

MOM: White Noise Variance

- For a general $MA(q)$ process, we recall that

$$\gamma_0 = (1 + \theta_1^2 + \theta_2^2 + \cdots + \theta_q^2)\sigma_e^2 \Rightarrow \sigma_e^2 = \frac{\gamma_0}{1 + \theta_1^2 + \theta_2^2 + \cdots + \theta_q^2}.$$

- Therefore, the MOM estimator of σ_e^2 is obtained by substituting in $\hat{\theta}_k$ for θ_k and S^2 for γ_0 . We obtain

$$\hat{\sigma}_e^2 = \frac{S^2}{1 + \hat{\theta}_1^2 + \hat{\theta}_2^2 + \cdots + \hat{\theta}_q^2}.$$

LEAST SQUARES ESTIMATION

Least Squares Estimation

- ▶ The MOM approach to estimation in stationary *ARMA* models is not always satisfactory.
- ▶ In fact, it is recommended to avoid MOM estimation in any model with moving average components.
- ▶ We, therefore, consider other estimation approaches, starting with conditional least squares (CLS).

Least Squares Estimation: $AR(1)$ Model

- ▶ Consider the stationary $AR(1)$ model

$$Y_t - \mu = \phi(Y_{t-1} - \mu) + e_t,$$

where a nonzero mean $\mu = E(Y_t)$ has been added for flexibility.

- ▶ For this model, the conditional sum of squares function is

$$SC(\phi, \mu) = \sum_{t=2}^n [(Y_t - \mu) - \phi(Y_{t-1} - \mu)]^2.$$

- ▶ With a sample of time series data $Y_1, Y_2, \dots, Y_n$, note that the $t = 1$ term does not make sense because there is no Y_0 observation.

Least Squares Estimation: $AR(1)$ Model

- ▶ The principle of least squares says to choose the values of ϕ and μ that will minimize $SC(\phi, \mu)$. For the $AR(1)$ model, this amounts to solving

$$\frac{\partial SC(\phi, \mu)}{\partial \phi} = 0, \quad \frac{\partial SC(\phi, \mu)}{\partial \mu} = 0$$

for ϕ and μ .

- ▶ In the $AR(1)$ model, the CLS estimators are

$$\hat{\phi} = \frac{\sum_{t=2}^n (Y_t - \bar{Y})(Y_{t-1} - \bar{Y})}{\sum_{t=2}^n (Y_t - \bar{Y})^2}, \quad \hat{\mu} \approx \bar{Y}.$$

Least Squares Estimation: $AR(p)$ Model

- In the general $AR(p)$ model, the conditional sum of squares function is (a function of $p + 1$ parameters):

$$SC(\phi_1, \phi_2, \dots, \phi_p, \mu) = \sum_{t=p+1}^n [(Y_t - \mu) - \phi_1(Y_{t-1} - \mu) - \phi_2(Y_{t-2} - \mu) - \dots - \phi_p(Y_{t-p} - \mu)]^2,$$

- The CLS estimators $\phi_1, \phi_2, \dots, \phi_p$ and μ are chosen to minimize $SC(\phi_1, \phi_2, \dots, \phi_p, \mu)$.

Least Squares Estimation: $AR(p)$ Model

- ▶ The CLS estimator of μ is

$$\hat{\mu} \approx \bar{Y},$$

an approximation when n is large (i.e., much larger than p). The CLS estimators for $\phi_1, \phi_2, \dots, \phi_p$ are well approximated by the solutions to the sample Yule-Walker equations.

- ▶ Therefore, in stationary AR models, the MOM and CLS estimates should be approximately equal.

Least Squares Estimation: $MA(q)$ Models

- Estimation via CLS can be carried out for any higher-order $MA(q)$ model in the same fashion. When $q > 1$, the problem becomes finding the values of $\theta_1, \theta_2, \dots, \theta_q$ such that

$$SC(\theta_1, \theta_2, \dots, \theta_q) = \sum_{t=1}^n e_t^2 = \sum_{t=1}^n (Y_t + \theta_1 e_{t-1} + \theta_2 e_{t-2} + \dots + \theta_q e_{t-q})^2,$$

is minimized, subject to the initial conditions that $e_0 = e_{-1} = \dots = e_{-q} = 0$.

- This can be done numerically by searching over all possible values of $\theta_1, \theta_2, \dots, \theta_q$ which yield an invertible solution.

MAXIMUM LIKELIHOOD ESTIMATION (MLE)

Maximum Likelihood Estimation (MLE)

Terminology:

- ▶ MLE is the most commonly used method to estimate unknown parameters.
- ▶ Advantages include using the entire sample and having good large-sample properties.
- ▶ No concerns about "start-up" values in time series models.

Maximum Likelihood Estimation (MLE)

- ▶ The likelihood function L describes the joint distribution of the data $Y_1, Y_2, \dots, Y_n$.
- ▶ When maximizing L with respect to model parameters, we find values that are most consistent with the observed data.
- ▶ Main disadvantage: Requires specifying a joint probability distribution for the random variables.

AR(1) Model

- ▶ To illustrate how maximum likelihood estimates are obtained, consider the $AR(1)$ model where have a sample size n :

$$Y_t - \mu = \phi(Y_{t-1} - \mu) + e_t,$$

where $\{e_t\}$ is a normal zero-mean white noise process with variance σ_e^2 :

$$e_t \sim N(0, \sigma_e^2).$$

- ▶ The parameters of the model are:
 - Autoregressive coefficient: ϕ
 - Mean: μ
 - Noise variance: σ_e^2

Joint PDF of $e_2, e_3, \dots, e_n$

- ▶ The probability density function (pdf) of e_t is assumed to be Gaussian:

$$f(e_t) = \frac{1}{\sqrt{2\pi\sigma_e^2}} \exp\left(-\frac{e_t^2}{2\sigma_e^2}\right), \quad \forall -\infty < e_t < \infty.$$

- ▶ Since $e_1, e_2, \dots, e_n$ are independent:

$$f(e_2, e_3, \dots, e_n) = \prod_{t=2}^n f(e_t) = (2\pi\sigma_e^2)^{-\frac{(n-1)}{2}} \exp\left(-\frac{1}{2\sigma_e^2} \sum_{t=2}^n e_t^2\right).$$

Joint PDF of Y

- Using transformations:

$$Y_2 = \mu + \phi(Y_1 - \mu) + e_2,$$

$$Y_3 = \mu + \phi(Y_2 - \mu) + e_3,$$

$$\vdots$$

$$Y_n = \mu + \phi(Y_{n-1} - \mu) + e_n.$$

- Y_1 is fixed, so we obtain:

$$L(\phi, \mu, \sigma_e^2 | y) = f(y_2, y_3, \dots, y_n | y_1) f(y_1).$$

Likelihood Function

- The joint pdf is:

$$f(y_2, \dots, y_n | y_1) = (2\pi\sigma_e^2)^{-\frac{(n-1)}{2}} \exp \left(-\frac{1}{2\sigma_e^2} \sum_{t=2}^n [(y_t - \mu) - \phi(y_{t-1} - \mu)]^2 \right).$$

- Multiplying by $f(y_1)$, we get:

$$L(\phi, \mu, \sigma_e^2 | y) = (2\pi\sigma_e^2)^{-\frac{n}{2}} (1 - \phi^2)^{1/2} \exp \left(-\frac{S(\phi, \mu)}{2\sigma_e^2} \right),$$

where:

$$S(\phi, \mu) = (1 - \phi^2)(y_1 - \mu)^2 + \sum_{t=2}^n [(y_t - \mu) - \phi(y_{t-1} - \mu)]^2.$$

Asymptotic Normality of MLEs

- For a stationary $ARMA(p, q)$ process, the MLEs satisfy:

$$\sqrt{n}(\hat{\phi}_j - \phi_j) \xrightarrow{d} N(0, \sigma_{\hat{\phi}_j}^2), \quad j = 1, \dots, p,$$

$$\sqrt{n}(\hat{\theta}_k - \theta_k) \xrightarrow{d} N(0, \sigma_{\hat{\theta}_k}^2), \quad k = 1, \dots, q.$$

- Implication: MLEs are consistent and asymptotically normal.

Example

- ▶ We consider the Gota River discharge data and use R to fit an $MA(1)$ model:

$$Y_t = \mu + e_t - \theta e_{t-1}$$

using the method of maximum likelihood.

R Output

```
> arima(gota,order=c(0,0,1),method='ML')
```

Coefficients:

	ma1	intercept
	0.5350	535.0311
s.e.	0.0594	10.4300

sigma² estimated as 6957: log likelihood = -876.58,
aic = 1757.15

Example

- ▶ We compare the estimates from all three methods (MOM, CLS, and MLE) with the Gota River discharge data:

Method	$\hat{\mu}$	$\hat{\theta}$	$\hat{\sigma}_e^2$
MOM	535.464	-0.654	6624
CLS	534.720	-0.535	6973
MLE	535.031	-0.535	6957

- ▶ Note that the CLS and ML estimates of θ are identical (to three decimal places). The MOM estimate of θ is noticeably different.